

CDS 101 Disney Project

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Shows Each Column with examples

```
glimpse(disney)
```

```
## Rows: 1,450
## Columns: 12
## $ show_id      <chr> "s1", "s2", "s3", "s4", "s5", "s6", "s7", "s8", "s9", "s1~
## $ type         <chr> "Movie", "Movie", "Movie", "Movie", "TV Show", "Movie", "~
## $ title        <chr> "Duck the Halls: A Mickey Mouse Christmas Special", "Erne~
## $ director     <chr> "Alonso Ramirez Ramos, Dave Wasson", "John Cherry", "Kare~
## $ cast         <chr> "Chris Diamantopoulos, Tony Anselmo, Tress MacNeille, Bil~
## $ country      <chr> "", "", "United States", "", "", "United States", "", "Un~
## $ date_added   <chr> "November 26, 2021", "November 26, 2021", "November 26, 2~
## $ release_year <int> 2016, 1988, 2011, 2021, 2021, 2021, 2021, 2015, 2019, 200~
## $ rating       <chr> "TV-G", "PG", "TV-G", "TV-PG", "", "PG-13", "TV-14", "TV~
## $ duration     <chr> "23 min", "91 min", "23 min", "41 min", "1 Season", "94 m~
## $ listed_in    <chr> "Animation, Family", "Comedy", "Animation, Comedy, Family~
## $ description  <chr> "Join Mickey and the gang as they duck the halls!", "Sant~
```

Shows each Variable

```
summary(disney)
```

```
##      show_id      type      title      director
## Length:1450    Length:1450    Length:1450    Length:1450
## Class :character Class :character Class :character Class :character
## Mode  :character Mode  :character Mode  :character Mode  :character
##
##
##      cast      country      date_added      release_year
## Length:1450    Length:1450    Length:1450    Min.   :1928
## Class :character Class :character Class :character    1st Qu.:1999
## Mode  :character Mode  :character Mode  :character    Median :2011
##                                     Mean   :2003
##                                     3rd Qu.:2018
```

```
##
##      rating      duration      listed_in      Max.      :2021
##      Length:1450      Length:1450      Length:1450      Length:1450
##      Class :character      Class :character      Class :character      Class :character
##      Mode :character      Mode :character      Mode :character      Mode :character
##
##
##
```

2.)

```
### Converts the release year variable from character to a numeric value
disney <- disney %>%
  mutate(release_year = as.numeric(release_year))
```

```
### Converts the duration variable from character to a numeric value
disney <- disney %>%
  mutate(duration_num = parse_number(duration))
```

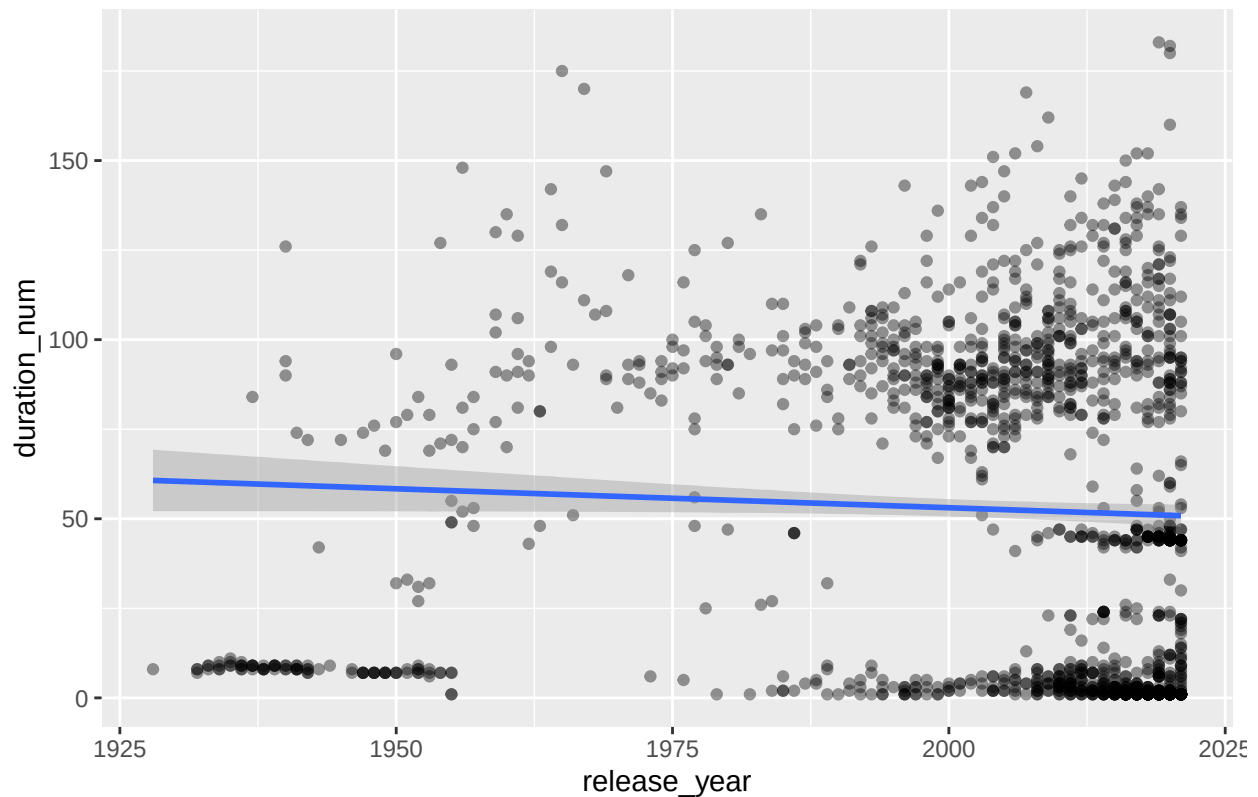
```
### Transforms the ratings and listed_in variables into factors which makes them categorical
disney <- disney %>%
  mutate(
    rating = as.factor(rating),
    listed_in = as.factor(listed_in)
  )
```

3.) Exploring Relationships:

```
#Creates a scatter plot looking at the relationship between the release year
#and the duration of the movie to see if there is any correlation between the two.
ggplot(disney, aes(x = release_year, y = duration_num)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm") +
  labs(title = "Duration vs Year of Release")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Duration vs Year of Release



ii.)

*# Adds a new column named period which splits release years into different periods
of time being 25 years apart from each other.*

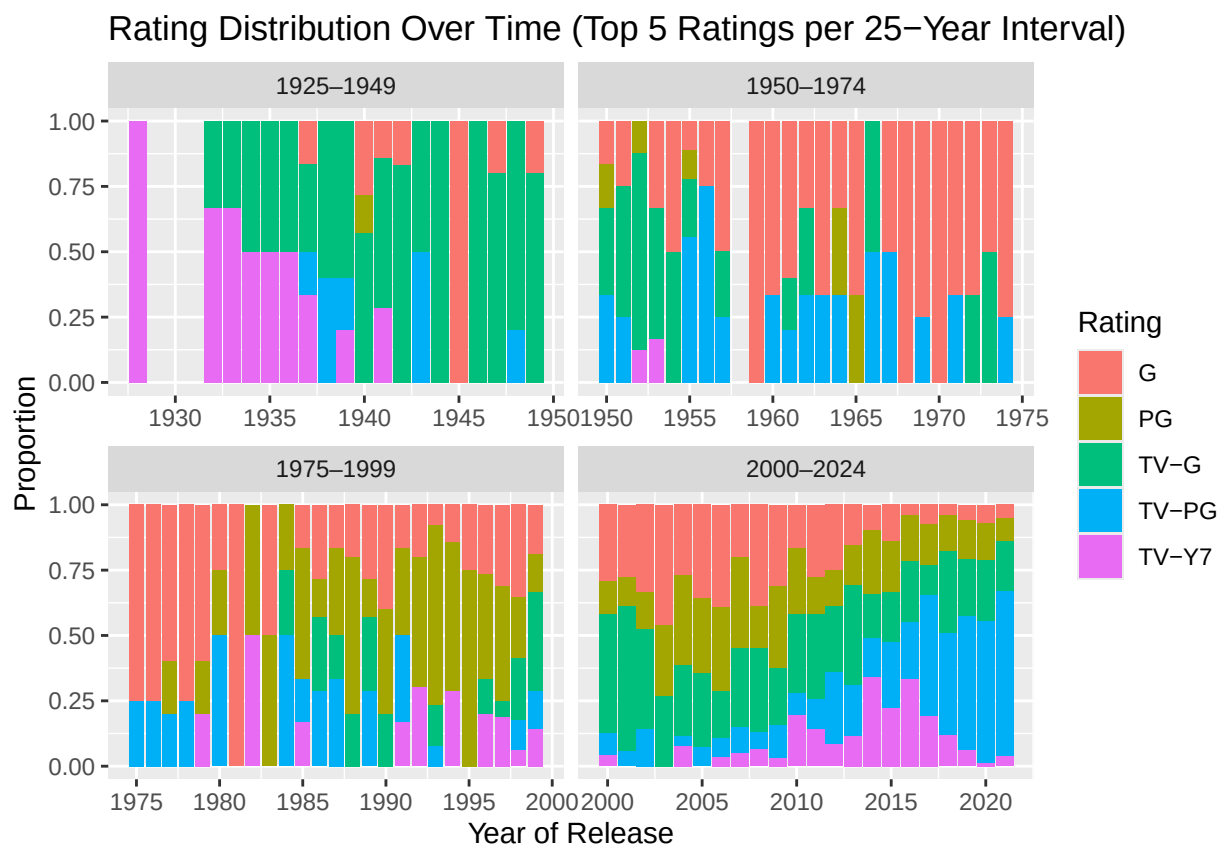
```
disney <- disney %>%
  mutate(period = cut(
    release_year,
    breaks = c(1900, 1925, 1950, 1975, 2000, 2025),
    labels = c("1900-1924", "1925-1949", "1950-1974", "1975-1999", "2000-2024"),
    right = FALSE
  ))
```

*# Creates a new data set looking at the amount of movies spread based on
two different factors.
The factors being time period and rating of the movie.
For example, This just gives the total amount of movies for rated G during
the years of 2000-2024*

```
top_ratings <- disney %>%
  count(period, rating, sort = TRUE) %>%
  group_by(period) %>%
  slice_max(n, n = 5) %>%
  ungroup()
```

```
#New dataset making sure all the values match from top_ratings using the semi_join
#function
ratings_top <- disney %>%
  semi_join(top_ratings, by = c("period", "rating"))
```

```
#Creates a bar plot looking at the distribution of age rating throughout the years.
ratings_top %>%
  count(period, release_year, rating) %>%
  ggplot(aes(x = release_year, y = n, fill = rating)) +
  geom_col(position = "fill") +
  facet_wrap(~ period, scales = "free_x") +
  labs(
    title = "Rating Distribution Over Time (Top 5 Ratings per 25-Year Interval)",
    x = "Year of Release",
    y = "Proportion",
    fill = "Rating"
  )
)
```



iii.)

```
# Creates a new data set with new rows based on the listed_in Variable.
# This new data set creates new rows based on the different genres of each movie.
```

```

# For example the first movie in the data set
# "Duck the Halls: A Mickey Mouse Christmas Special"
# is split into two different two rows with one having animation and the other being family.
# Not all rows will be split, but all rows will keep the other same data
# such as rating and release year.
disney_genre <- disney %>%
  separate_rows(listed_in, sep = ", ") %>%
  mutate(listed_in = as.factor(listed_in))

```

```

# This does the same thing as earlier, adding a new column named period.
# This adds a column based on the release date of the movie,
# being 25 years apart from each other.
disney_genre <- disney_genre %>%
  mutate(period = cut(
    release_year,
    breaks = c(1900, 1925, 1950, 1975, 2000, 2025),
    labels = c("1900-1924", "1925-1949", "1950-1974", "1975-1999", "2000-2024"),
    right = FALSE
  ))

```

```

# Does the same thing earlier and counts the total amount of movies
# based on rating sorted year
top_genres <- disney_genre %>%
  count(period, listed_in, sort = TRUE) %>%
  group_by(period) %>%
  slice_max(n, n = 5) %>%
  ungroup()

```

```

#New dataset making sure all the values match from disney_genre using the semi_join
#function
disney_top <- disney_genre %>%
  semi_join(top_genres, by = c("period", "listed_in"))

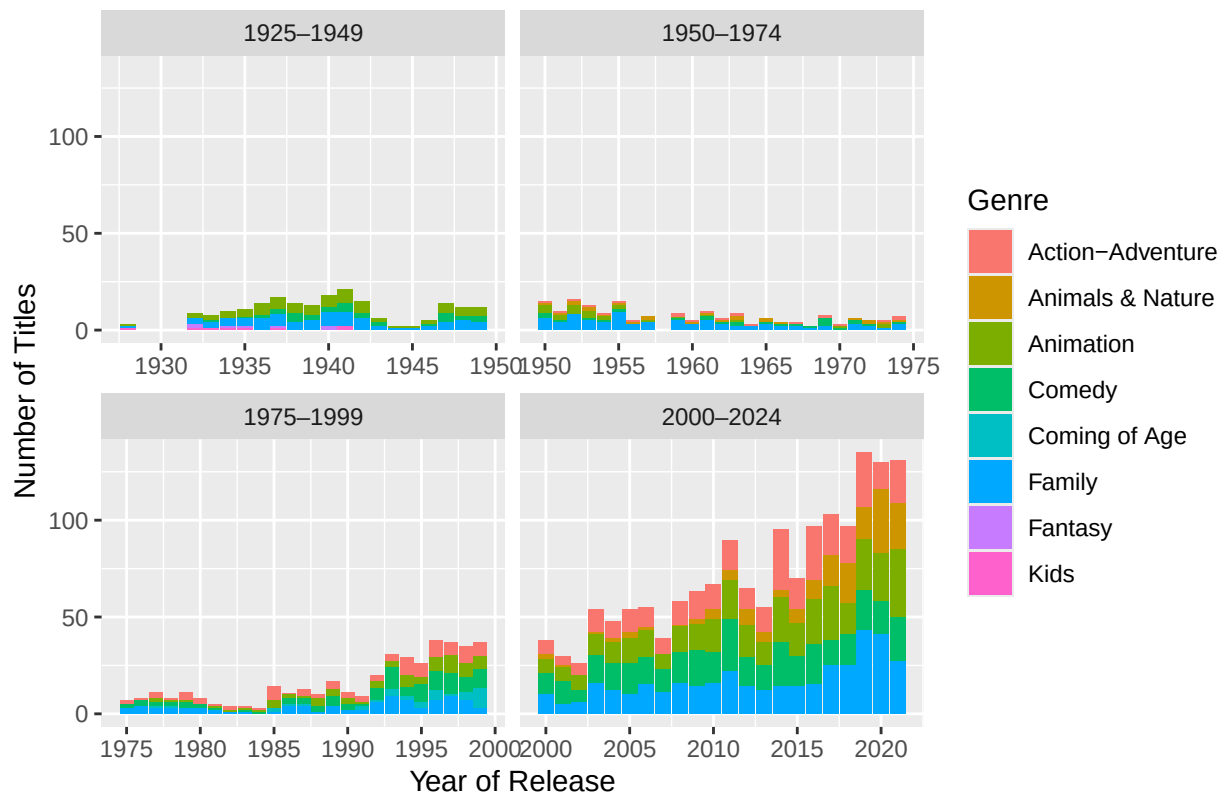
```

```

# Creates multiple bar plots looking at the top 5 genres
# at a with a range of 25 years.
# Shows the relationship between genres and intervalss of release.
disney_top %>%
  count(period, release_year, listed_in) %>%
  ggplot(aes(x = release_year, y = n, fill = listed_in)) +
  geom_col() +
  facet_wrap(~ period, scales = "free_x") +
  labs(
    title = "Top 5 Genres per 25-Year Interval",
    x = "Year of Release",
    y = "Number of Titles",
    fill = "Genre"
  )

```

Top 5 Genres per 25-Year Interval



4.)

```
# fits the duration and release year into a linear regression
# model and provides statistics such as p values, estimates,
# r.squared, and sigma
model_duration <- lm(duration_num ~ release_year, data = disney)
tidy(model_duration)
glance(model_duration)
```

```
# Fits the rating year into a linear regression model
# based on the release years and provides the same statistics
model_rating <- lm(as.numeric(rating) ~ release_year, data = disney)
tidy(model_rating)
```

5.)

```
# Does the same thing as the other code chunks, basically asking
# How release year, rating, and genre impact the movie duration
model_multi <- lm(duration_num ~ release_year + rating + listed_in, data = disney)
tidy(model_multi)
glance(model_multi)
```

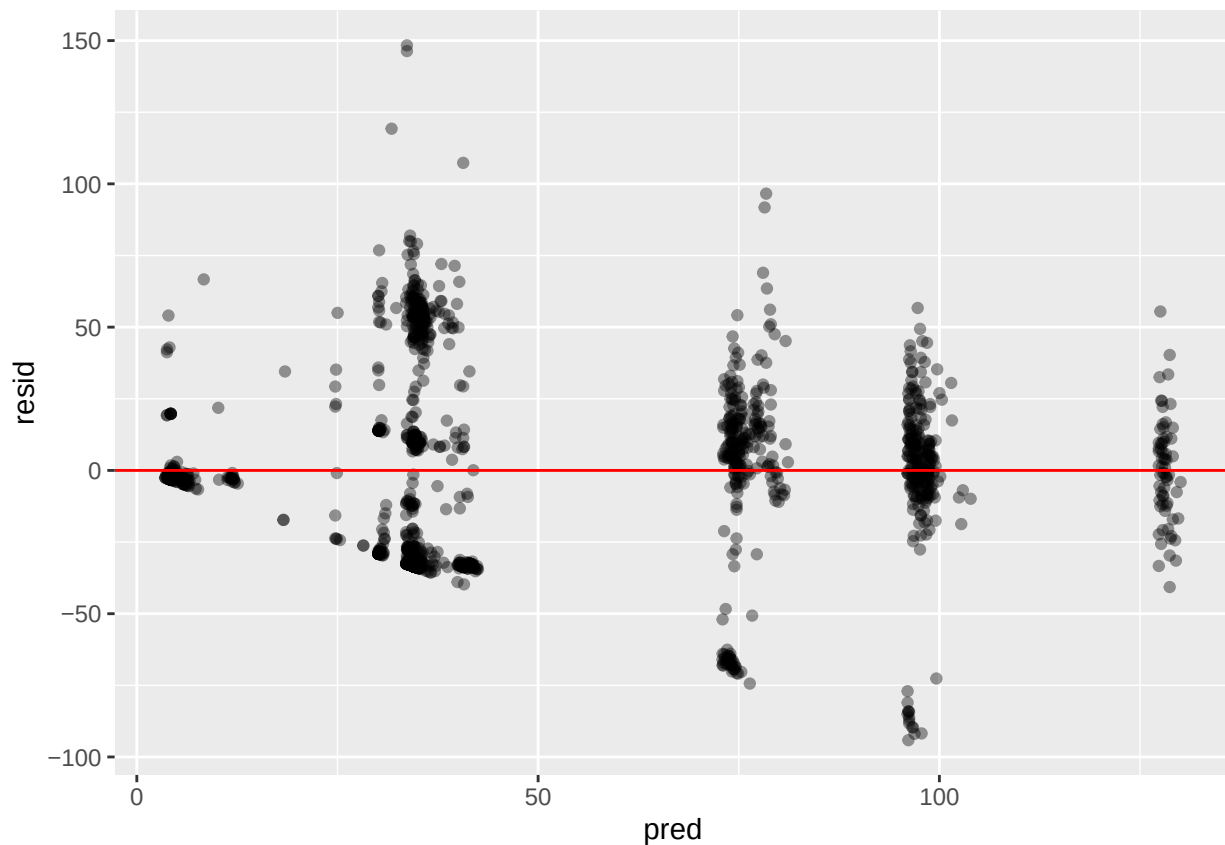
```
# This creates a linear regression model looking and how
# only release year and rating predict duration.
# and provides statistics like previous code chunks
model_multi1 <- lm(duration_num ~ release_year + rating, data = disney)
tidy(model_multi1)
glance(model_multi1)
```

6.)

```
# Adds the pred and resid variables from the previous
# code chunks to a new dataset named disney_pred
disney_pred <- disney %>%
  add_predictions(model_multi1, var = "pred") %>%
  add_residuals(model_multi1, var = "resid")
```

7.)

```
# Plots residual vs predicted durations
# x= predicted duration
# y = residuals (actual - predicted)
ggplot(disney_pred, aes(x = pred, y = resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, color = "red")
```



```

# Plots predicted vs actual duration
# X is the predicted duration
# y is the acutal duration of the movie
ggplot(disney_pred, aes(x = pred, y = duration_num)) +
  geom_point(alpha = 0.4) +
  geom_abline(color = "blue")

```

